

Image Reconstruction via Compressed Sensing

Overview

This project will implement and evaluate compressed sensing techniques for image reconstruction from limited measurements. Compressed sensing allows recovery of signals from far fewer samples than traditional methods require, making it valuable for applications like MRI, satellite imaging, and digital photography.

Objectives

The project will focus on a simplified 2D grayscale image reconstruction problem. More specifically, it includes:

1. Implementation of image reconstruction algorithms using optimization techniques
2. Comparison of the performance of different algorithms on simple test images
3. Analysis of the trade-offs between measurement count and reconstruction quality

Methodology

1. Data Generation

- Create synthetic measurements of test images (64×64 or 128×128 pixels)
- Apply random Gaussian or partial Fourier measurement matrices
- Generate measurements at various sampling rates (10%, 25%, 50%)

2. Algorithms to implement

- LASSO formulation: $\min_x \frac{1}{2} \|Ax - y\|_2^2 + \lambda \|x\|_1$
- Implementation of two algorithms: (1) Proximal Gradient Descent (ISTA) and (2) Fast Iterative Shrinkage-Thresholding Algorithm (FISTA).

3. Evaluation Metrics

- Peak Signal-to-Noise Ratio (PSNR)
- Reconstruction time
- Convergence behavior (objective function vs. iterations)

Deliverables

Deliverables will include the final report documenting:

- Mathematical formulation
- Implementation details
- Experimental results with visualizations
- Discussion of findings and potential improvements